

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

ROTATIONAL MOTION

LECTURE-4

PHYSICS

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Topics To Be Covered

- 1 Revision ✓
- 2 Torque ✓
- 3 Direction of Torque ✓
- 4 Relationship between Torque and Angular Momentum




- 5 .. Moment of Inertia ($I = mR^2$)

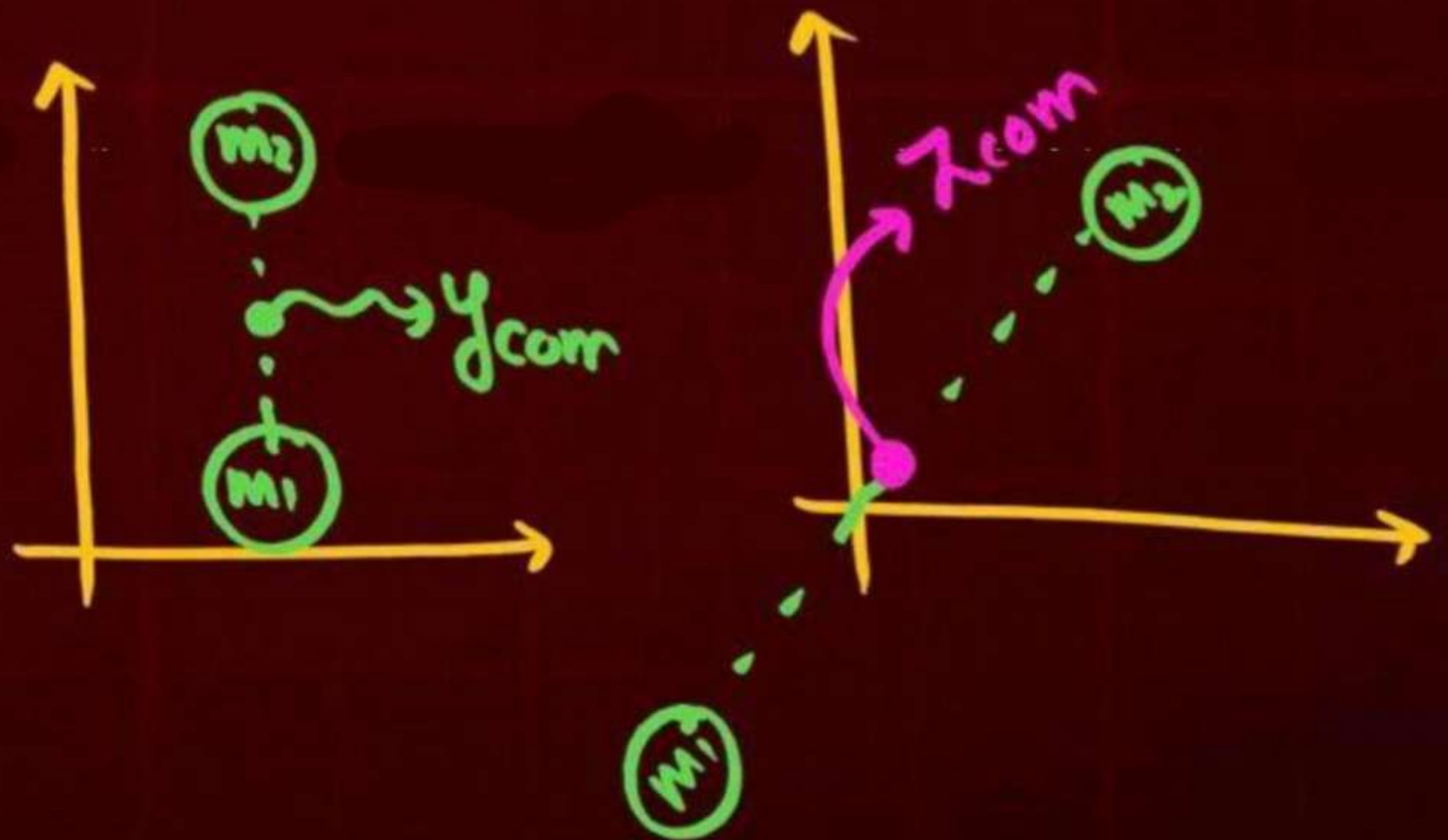
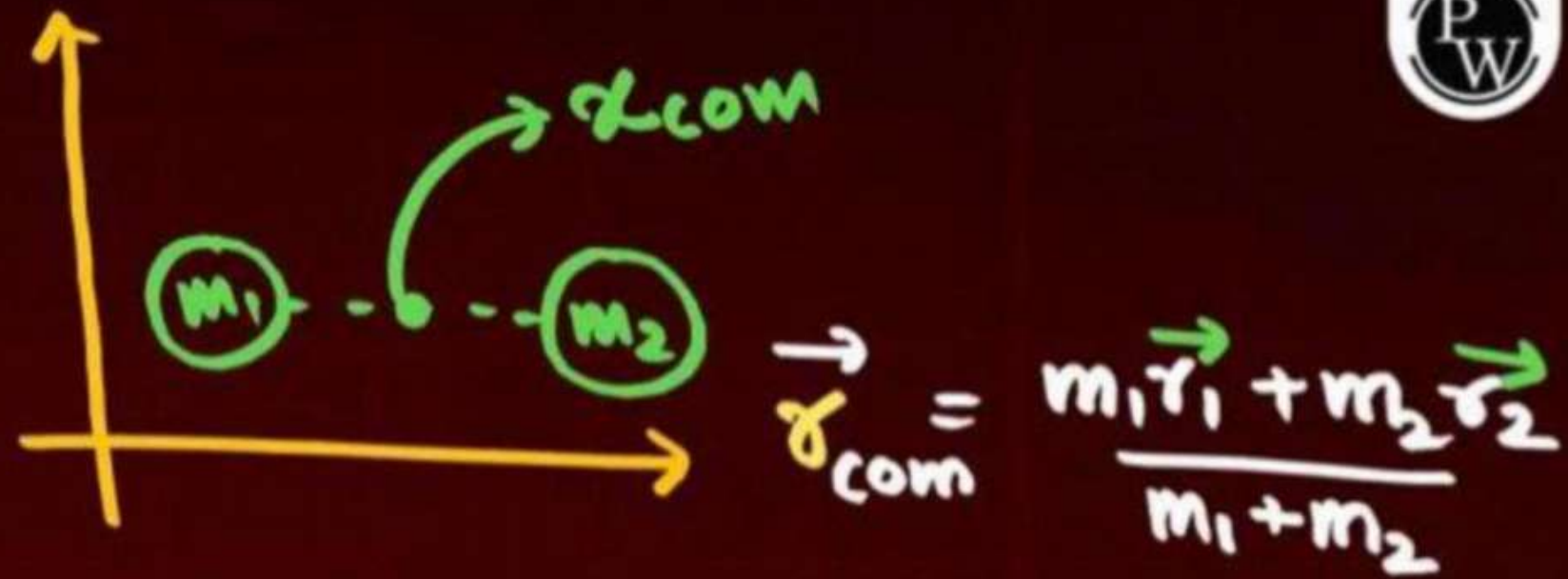
Recap From 1st Lecture

Position of Com

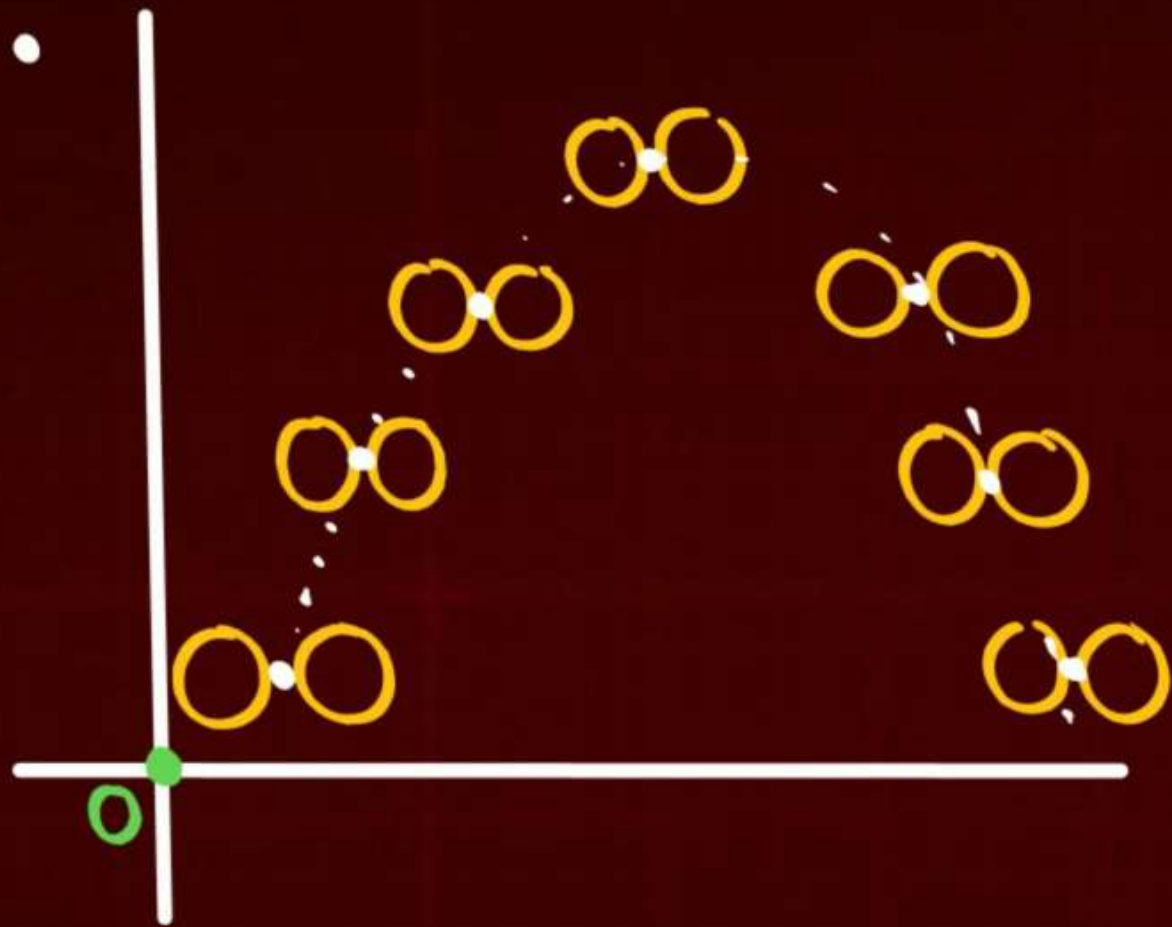
$$x_{com} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$y_{com} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$z_{com} = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2}$$



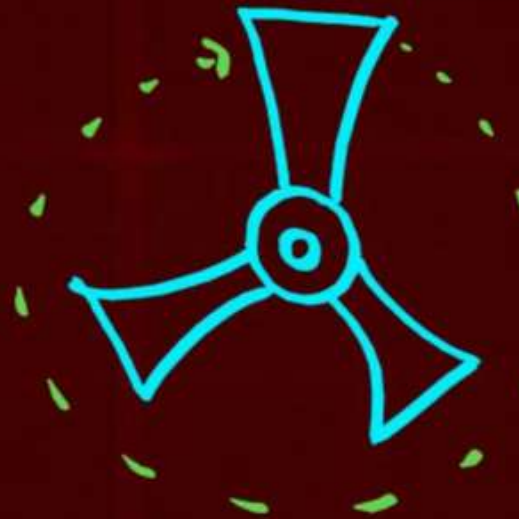
RECAP FROM 2nd LECTURE



$$\textcircled{1} V_{\text{com}} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$\textcircled{2} a_{\text{com}} = \frac{m_1 a_1 + m_2 a_2}{m_1 + m_2}$$

Rotational Motion



✓ Angular displacement

$$\theta = \frac{\text{Arc}}{R} \quad x = R\theta \quad \text{Rad}$$

✓ Angular velocity

$$\omega = \frac{\theta}{t} \quad v = R\omega \quad \text{Rad/sec}$$

✓ Angular acceleration

$$\alpha = \frac{\omega}{t} \quad a = R\alpha \quad \text{Rad/sec}^2$$

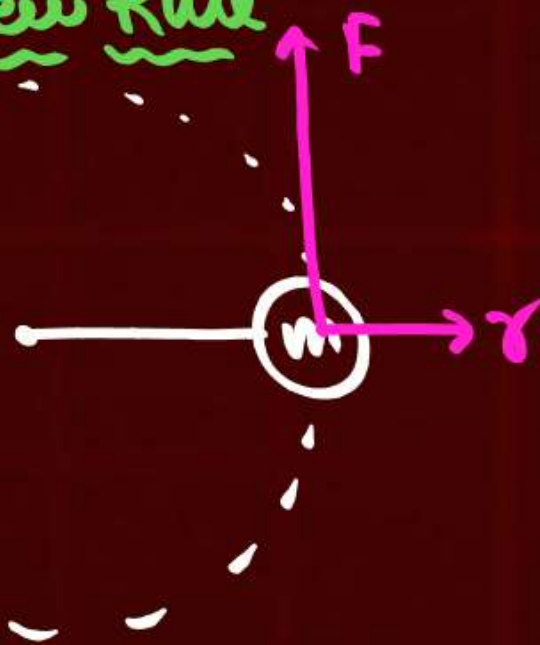
Recap From 3rd Lecture

Angular momentum

$$\vec{L} = \vec{r} \times \vec{p}$$

$$L = r p \sin \theta \hat{n}$$

Screw Rule



Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = r F \sin \theta \hat{n}$$

numerical

$$\checkmark \vec{r} = 2\hat{i} + 3\hat{j} \quad \vec{p} = 3\hat{i} + 4\hat{j}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\checkmark \vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k} \quad \vec{p} = 3\hat{i} + 4\hat{j} + 3\hat{k}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

Trick

lecture no. 4

Torque $\vec{\tau} \Rightarrow \text{Tau}$

- The Rotating effect of force about the axis of rotation is called

Torque

OR

The Torque is a cross product of $\vec{r}$ and $\vec{F}$

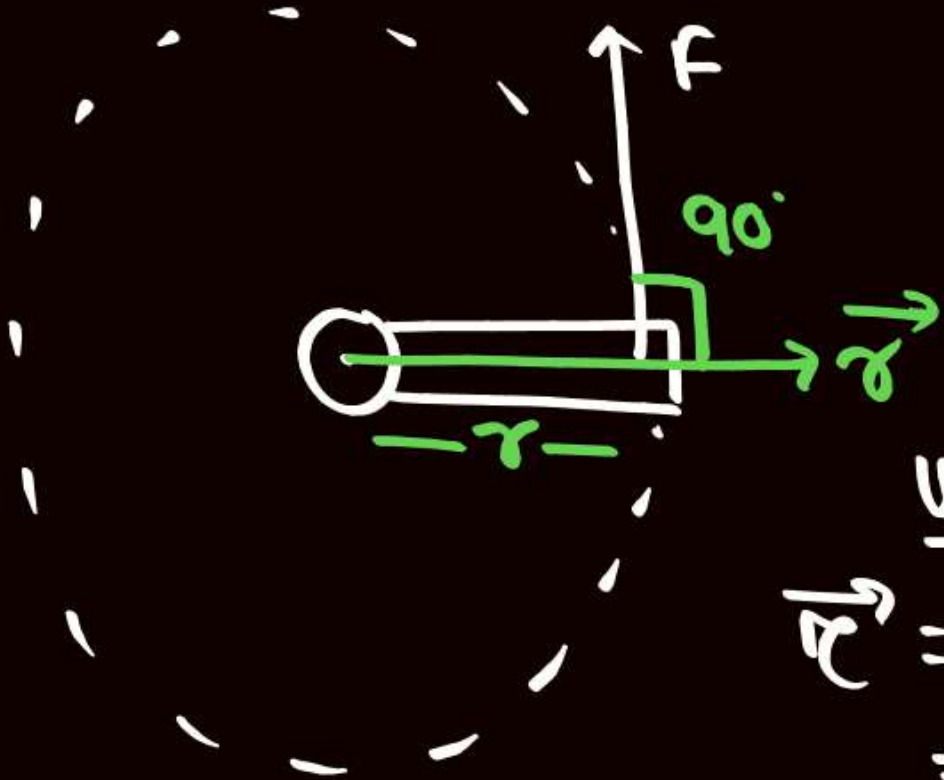
$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = r F \sin \theta \hat{n}$$

direction

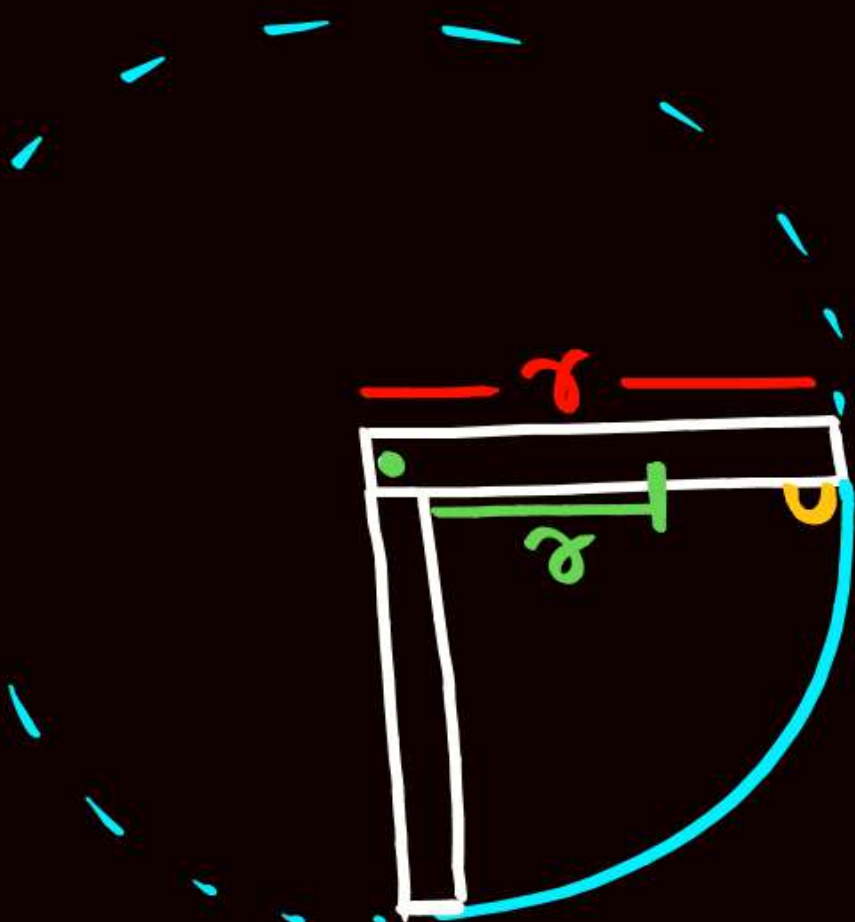
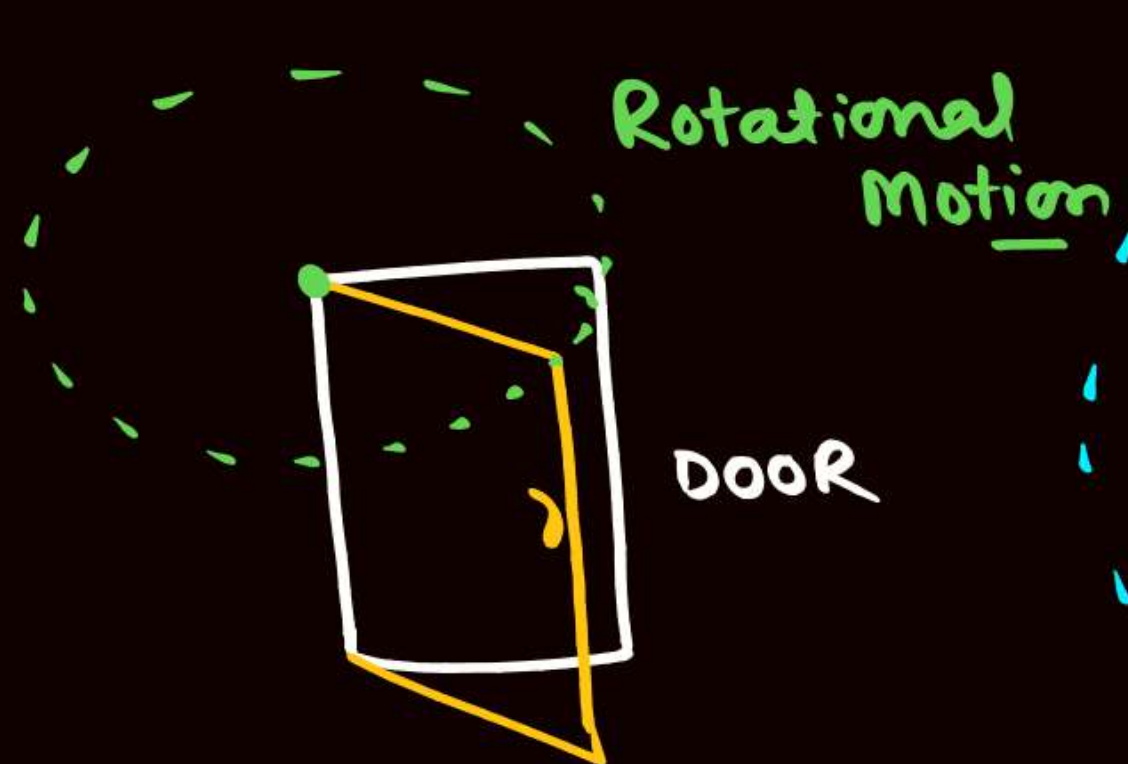


Screw Rule



$$\begin{aligned} \tau &= \text{unit} \\ &= \text{Nm} \\ &= \text{Nm} \end{aligned}$$

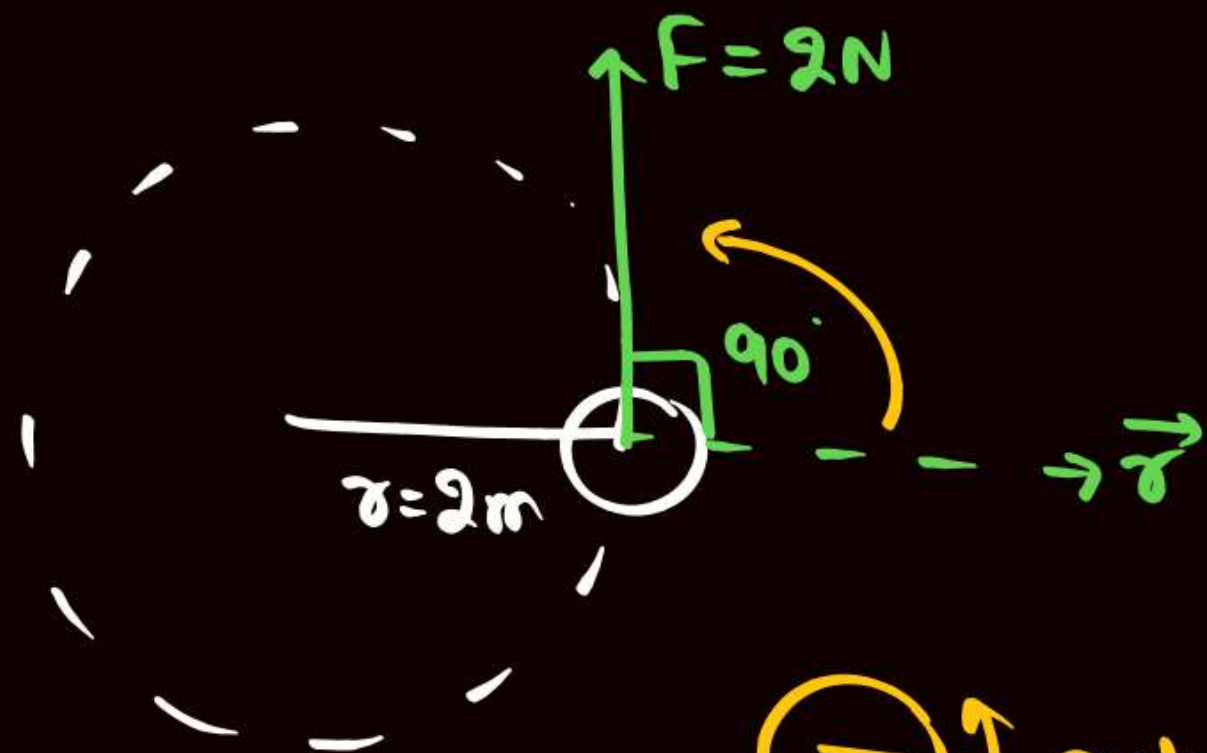
Example



$$\vec{\tau} = \vec{r} \times \vec{F}$$

Diagram illustrating the cross product $\vec{\tau} = \vec{r} \times \vec{F}$. The first part shows a vector $\vec{r}$ pointing up and a vector $\vec{F}$ pointing down, resulting in a torque vector $\vec{\tau}$ pointing out of the page (indicated by a green arrow pointing up). The second part shows a vector $\vec{r}$ pointing down and a vector $\vec{F}$ pointing up, resulting in a torque vector $\vec{\tau}$ pointing into the page (indicated by a green arrow pointing down).

Numerical ①



$$\tau = ?$$

Sol

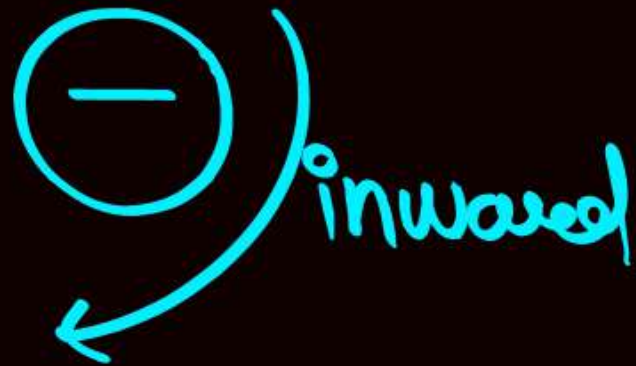
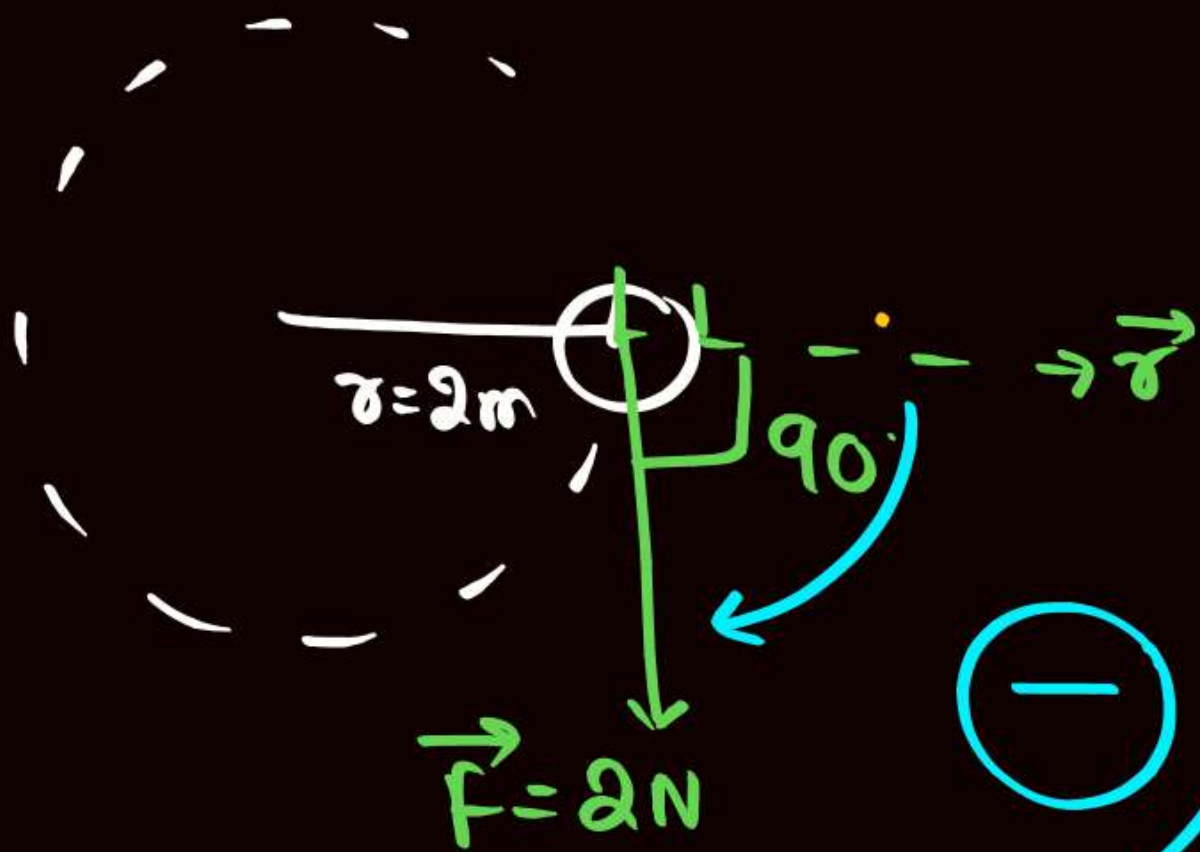
$$\begin{aligned}\vec{\tau} &= \vec{r} \times \vec{F} \\ &= r F \sin \theta \hat{n} \\ &= 2 \times 2 \sin 90^\circ \hat{n} \\ &= 4 \times 1 \hat{n} \\ &= 4 \hat{n} \\ &= 4 \hat{k} \text{ Nm}\end{aligned}$$

⊙ outward

Numerical 1

$$\tau = ?$$

Sol



$$\begin{aligned}\vec{\tau} &= \vec{r} \times \vec{F} \\ &= r F \sin \theta \hat{n} \\ &= 2 \times 2 \sin 90^\circ \hat{n} \\ &= 4 \times 1 \times \hat{n} \\ &= 4 \hat{n} \\ &= 4 \times -\hat{k} \\ &= -4 \hat{k} \text{ (Nm)}\end{aligned}$$

Neet - 2022

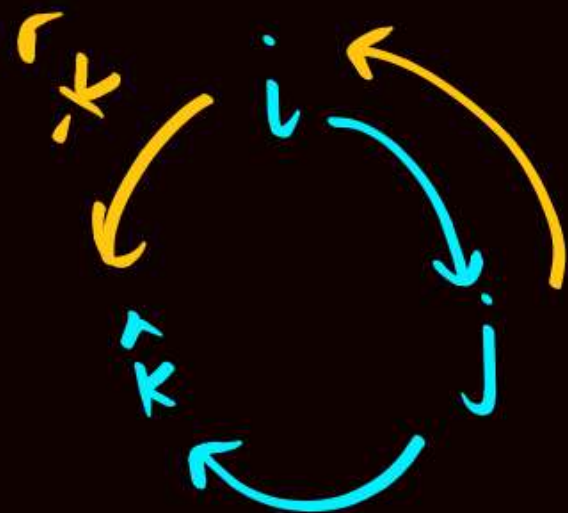
Q $\vec{r} = 2\hat{i} + 3\hat{j}$
 $\vec{F} = 2\hat{i} + 5\hat{j}$

$\vec{\tau} = ?$

$\Rightarrow \theta = 0$



$\Rightarrow \theta = 0$



Sol $\vec{\tau} = \vec{r} \times \vec{F}$
 $= (2\hat{i} + 3\hat{j}) \times (2\hat{i} + 5\hat{j})$

$= \underline{2\hat{i} \times 2\hat{i}} + \underline{2\hat{i} \times 5\hat{j}} + \underline{3\hat{j} \times 2\hat{i}} + \underline{3\hat{j} \times 5\hat{j}}$

$= \underline{2 \times 2 \sin 0^\circ} + 2 \times 5 \times \sin 90^\circ \hat{k} + 3 \times 2 \sin 90^\circ (-\hat{k}) + 3 \times 5 \sin 0^\circ$

$= 10 \times 1 \times \hat{k} + 6 \times 1 (-\hat{k})$

$= 10\hat{k} - 6\hat{k}$

$= \underline{4\hat{k}} \text{ (Nm)} \checkmark$

H/W
Q

$$\vec{r} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\vec{r} = 4\hat{i} + 5\hat{j} + 6\hat{k}$$

$$\vec{r} = ?$$

JKBOSE-2019

Derive the Relationship Between Torque and Angular Momentum

"Torque is Rate of change of Angular momentum" JKROSE 2023

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Acc. to Newton 2nd law

$$\vec{F} = \frac{d\vec{p}}{dt}$$

on multiplying b/s by $\vec{r}$

$$\vec{r} \times \vec{F} = \vec{r} \times \frac{d\vec{p}}{dt}$$

$$\vec{r} \times \vec{F} = \frac{d(\vec{r} \times \vec{p})}{dt}$$

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

$$\vec{r} \times \vec{F} = \vec{\tau}$$

$$\vec{r} \times \vec{p} = \vec{L}$$

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

Newton 2nd law in Rotation

Kar Ke Dikhao

The rate of change ^{↑ angular} momentum = 4
Calculate the Torque?

- (a) 4 Nm
- (b) 8 Nm
- (c) 16 Nm
- (d) 32 Nm

Torque is Rate of change of Angular momentum

$$\tau = R \text{ of change of } L$$

$$\tau = \underline{4 \text{ Nm}}$$

Equilibrium

Translational
Linear Motion



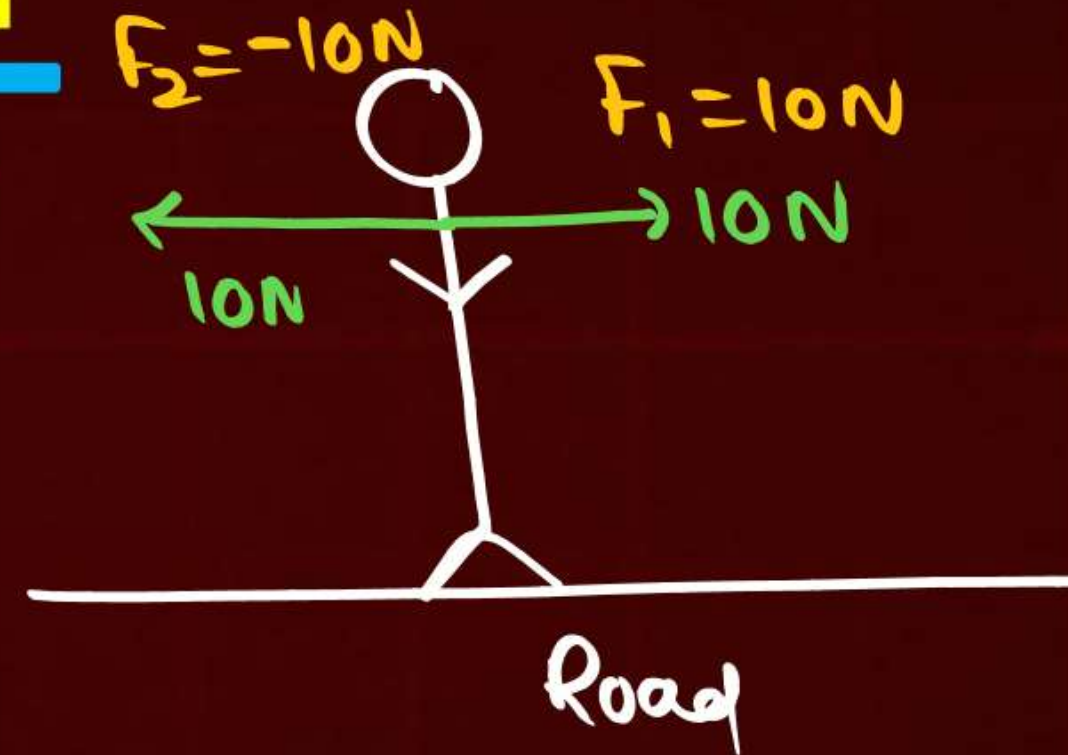
(i) Translational Equilibrium

when net force acting on a Body is Zero

$$\begin{aligned} F_{\text{net}} &= F_1 + F_2 \\ &= 10\text{N} + -10\text{N} \\ &= 10\text{N} - 10\text{N} \\ &= 0 \end{aligned}$$

⇓
the
Body
is

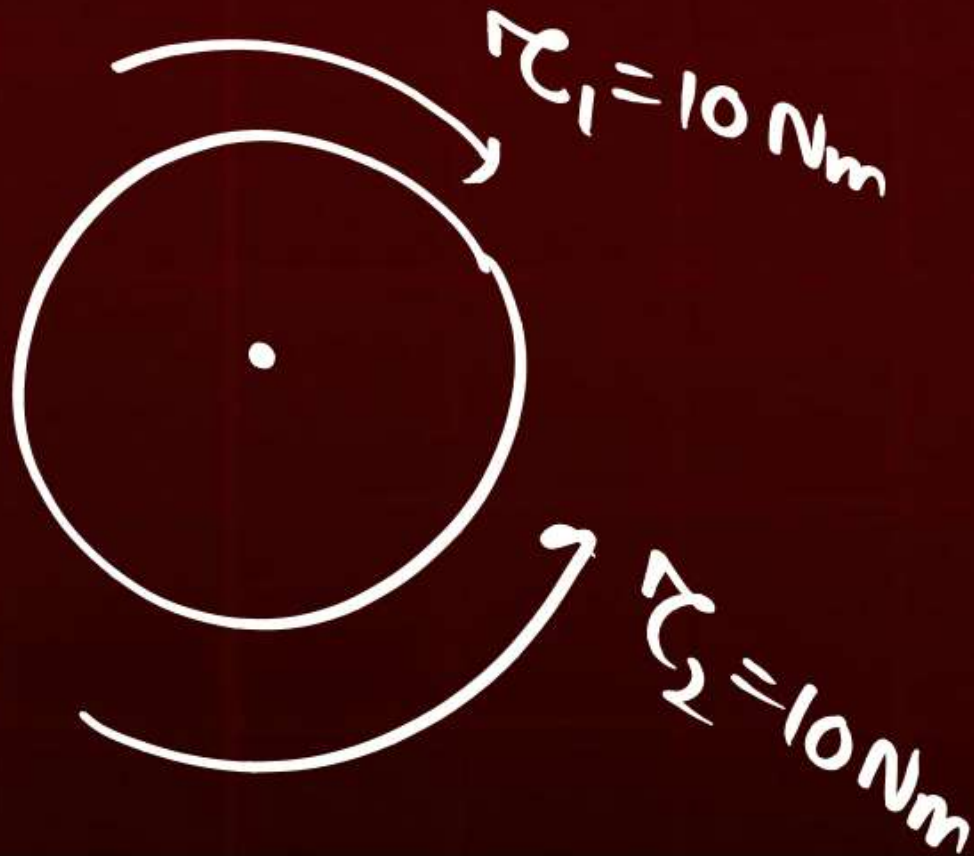
Said to be in Translational Equilibrium





② Rotational Equilibrium

When net Torque acting on a Body is Zero



$$\begin{aligned}\tau_{\text{net}} &= \tau_1 + \tau_2 \\ &= 10 \text{ Nm} + -10 \text{ Nm} \\ &= 10 \text{ Nm} - 10 \text{ Nm} \\ &= 0 \text{ Nm}\end{aligned}$$

Total equilibrium :- when net force and net Torque acting on a Body is Zero

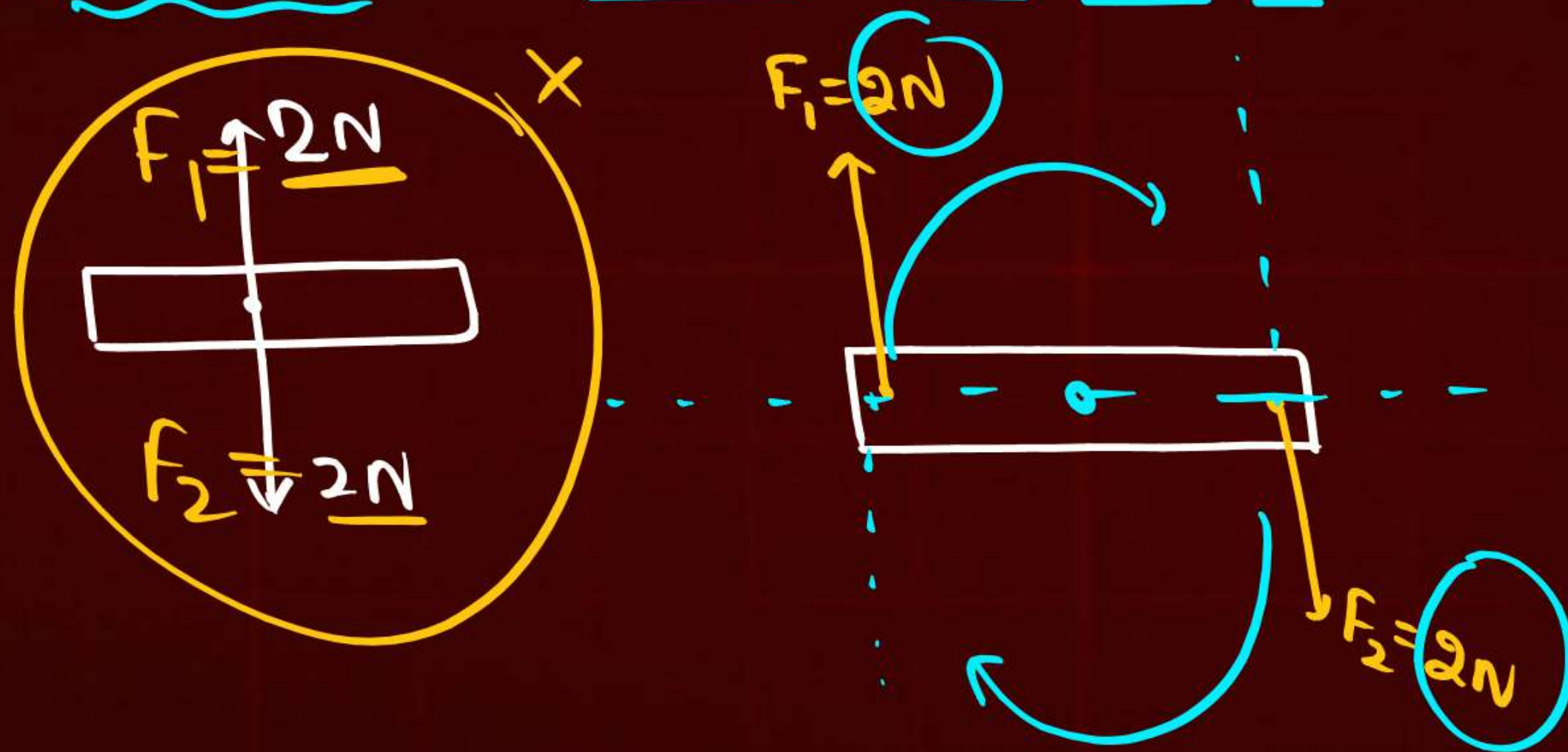
$$F_{\text{net}} = 0$$

$$\tau_{\text{net}} = 0$$

Couple

⇒ 12th

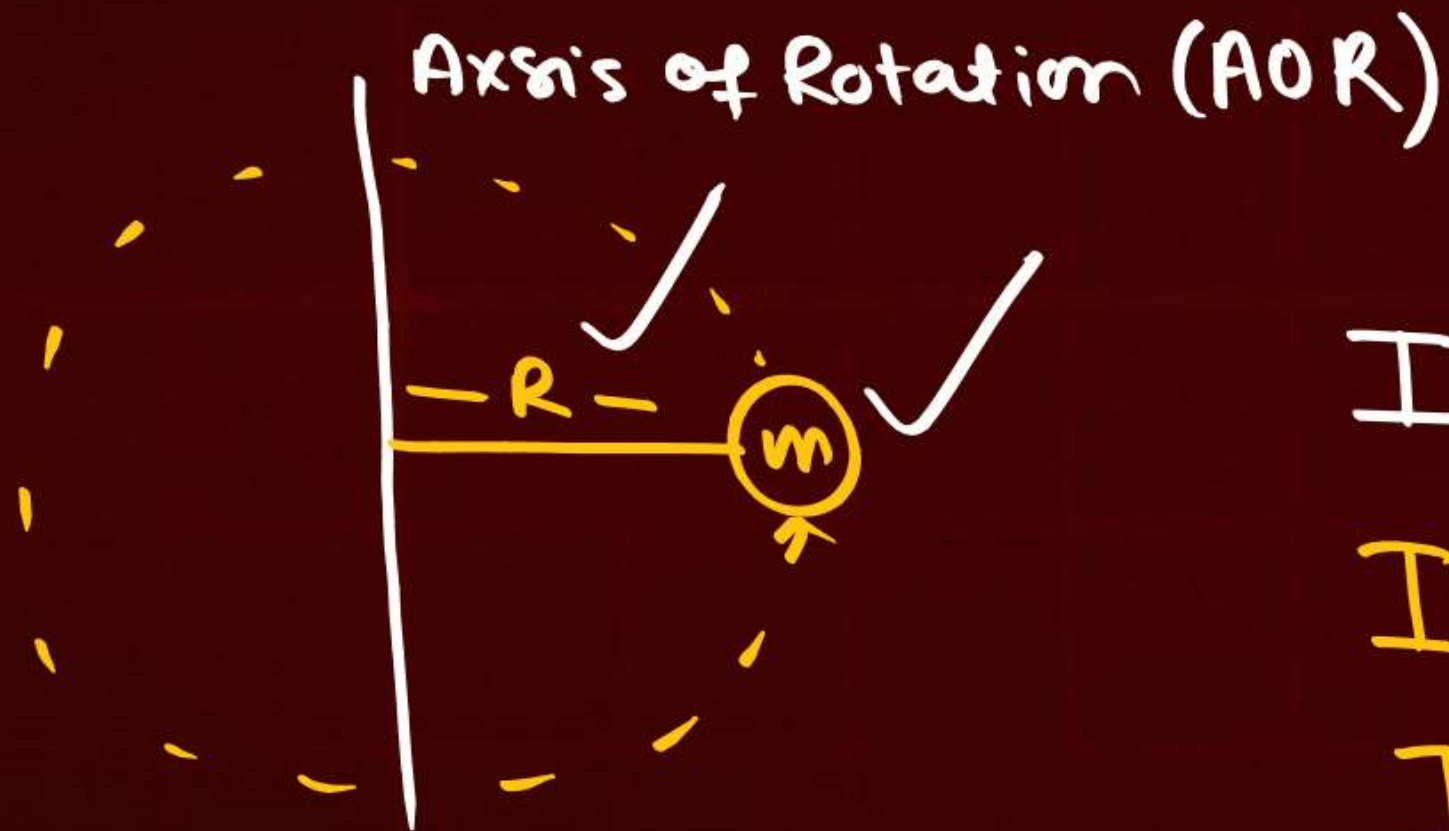
A couple or a force couple refers to two parallel forces that are equal in magnitude, opposite in sense and do not share a line of action.



Moment of inertia

$$I = mR^2$$

The moment of inertia is a measure of how resistant an object is to changes in its rotational motion.



$$I \propto m \text{ --- (I)}$$

$$I \propto R^2 \text{ --- (II)}$$

$$I \propto mR^2$$

$$I = K m R^2 \quad K = 1$$

$$I = 1 \times m R^2$$

$$\boxed{I = m R^2}$$

Neet = 2017

Q Dimensional formula of
 $I = ?$

Sol

$$\begin{aligned} I &= \underline{m} R^2 \\ &= m L^2 \\ &= \underline{m L^2} \end{aligned}$$

JKBOSE - mco 2024

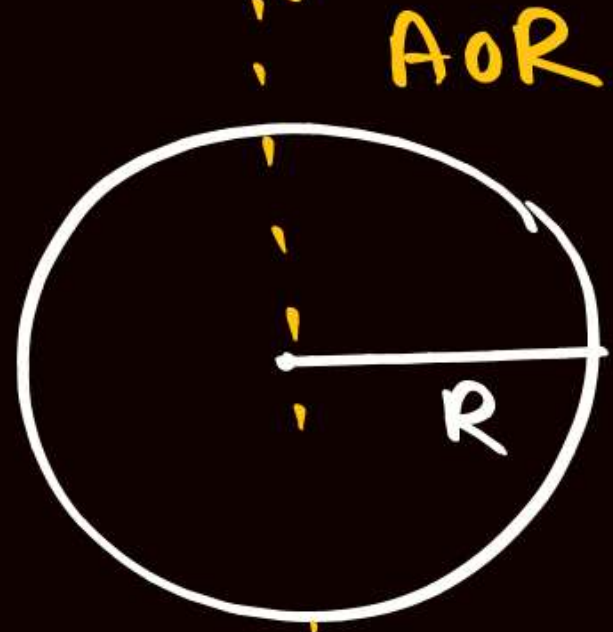
Q Give the unit of $\textcircled{I}$

Sol

$$\begin{aligned} I &= m R^2 \\ &= \text{kg m}^2 \\ &= \underline{\text{kg m}^2} \end{aligned}$$

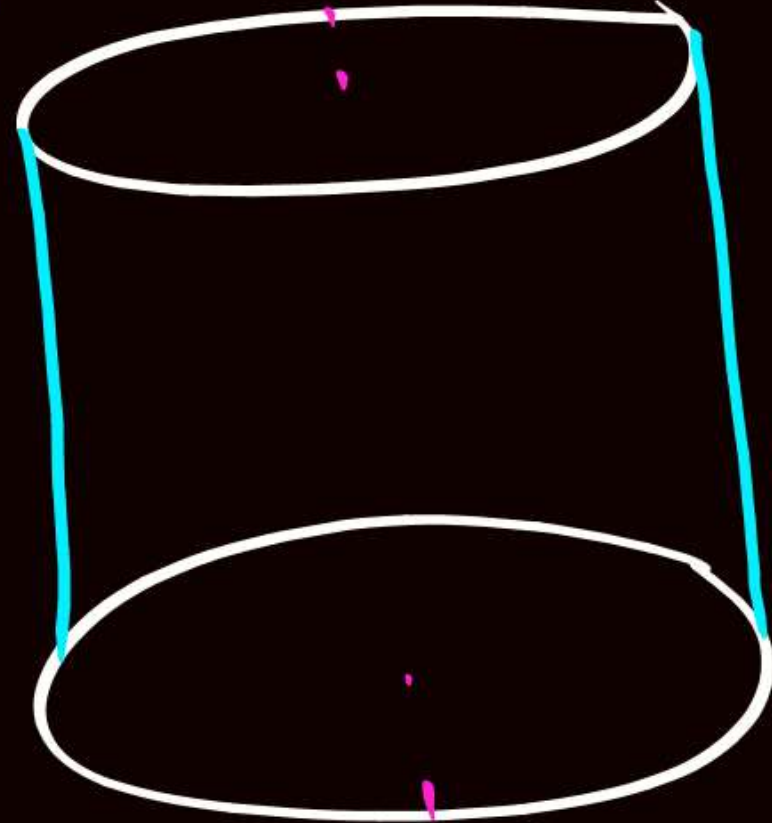
MOI of Rigid Body

(i) Ring



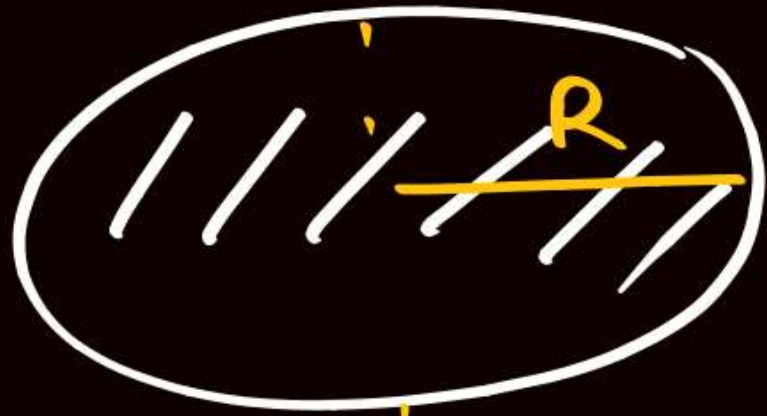
$$I = mR^2$$

(Hollow)
(ii) Cylinder



$$I = mR^2$$

(iii) Disc



$$I = \frac{1}{2} m R^2$$

Yaad

(iv) Solid Cylinder



$$I = \frac{1}{2} m R^2$$

(v) Rod (Middle)



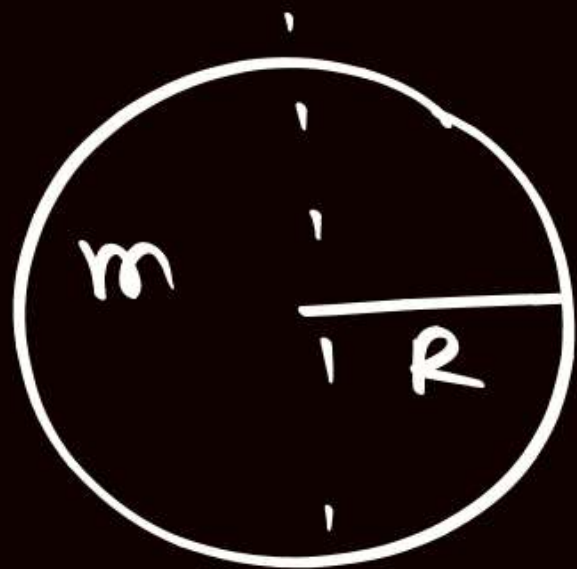
$$I_m = \frac{1}{12} ml^2$$

(vi) Rod (End)



$$I_e = \frac{1}{3} ml^2$$

(vii) Hollow sphere (ball)



$$I = \frac{2}{3} m R^2$$

(viii) Solid sphere (Ball)



$$I = \frac{2}{5} m R^2$$

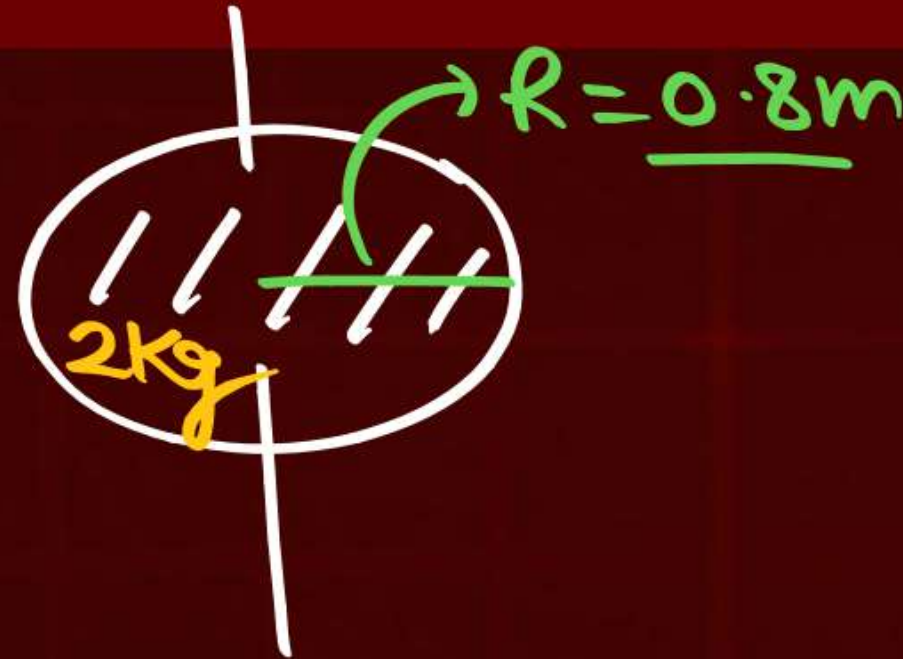
Ye Easy hai?

JKBOSE — 2017



Find MOI^I of a disc of radius 0.8 m and mass is 2 kg. When AOR is passed through perpendicular from center of disc.

Sol
 $I = ?$
 $R = 0.8\text{m}$
 $m = 2\text{kg}$



$$\begin{aligned} I &= \frac{1}{2} m R^2 \\ &= \frac{1}{2} \times 2 \times (0.8)^2 \\ &= 1 \times 0.8 \times 0.8 \\ &= 0.64 \text{ Kg m}^2 \end{aligned}$$

H/W
Q

$$\vec{r} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\vec{r} = 4\hat{i} + 5\hat{j} + 6\hat{k}$$

$$\vec{r} = ?$$

JKBOSE-2019

“TUM ITNA PADHNA KI JAB SAMNE
WALA TUMSE BAAT KARE TO
USKE PAISE KA ABHIMAN
TUMHARE GYAN KE SAMNE
CHHOTA PAD JAYE

Thank
You

